

Turning Chaos into Victory: Optimizing Random Team Composition in E-Sports Matches

Executive Summary

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Introduction

Esports analytics is a rapidly evolving field, leveraging data to gain a competitive edge in both professional and casual gaming (Gill, 2025). League of Legends, one of the most popular multiplayer online battle arena (MOBA) games, includes a unique game mode called All Random All Mid (ARAM). Unlike standard competitive play, where teams strategically select their champions, ARAM assigns champions at random (ARAM, 2025). This randomness creates a fascinating challenge: do certain team compositions inherently perform better, despite the lack of deliberate selection?

This study explores the impact of team composition on match outcomes in ARAM and investigates whether predictive models can identify patterns that improve decision-making. By analyzing match data and applying machine learning techniques, this research determines whether players can maximize their chances of victory through strategic use of re-rolls and champion trades. The findings provide data-driven insights to help players make informed champion selections to adapt to the unpredictability of ARAM.

Research Problem and Hypothesis

A fundamental assumption in ARAM is that the randomness of champion assignment results in a 50/50 win-loss distribution over the long term. However, player intuition suggests that certain team compositions outperform others. This research aims to test that intuition by analyzing thousands of matches to determine whether team composition alone can predict match outcomes.

- **Null Hypothesis (H_0):** A predictive model cannot use team composition to predict match outcomes better than random guessing ($\leq 50\%$ accuracy).
- **Alternative Hypothesis (H_1):** A predictive model can use team composition to predict match outcomes better than random guessing ($> 50\%$ accuracy).

By testing this hypothesis, this analysis assesses whether there are composition-based advantages in ARAM and how they can be leveraged for better pre-match champion selection and improving a team's probability of winning.

Data Collection and Analysis Process

Data Collection

To analyze match outcomes, data was collected from the game's API, which provides detailed records of ARAM matches, including champion selections, and match results. The initial dataset included 20,012 ARAM matches, each with two teams, resulting in 40,024 total team data points. During collection, the following filters were applied:

- Match duration ≥ 8 minutes
- Maximum of 20 matches per individual player
- Matches are from the most recent game version

Data Preparation:

The dataset underwent extensive preprocessing. Champion detail statistics and primary roles were merged into the dataset. Base statistics such as health, armor, movement speed, were dropped since they undergo significant changes during a match due to item purchases (*Item*, 2025). Champion primary roles (Assassin, Tank, Mage, Marksman, Support, Fighter) were one-hot encoded.

The champion roles and remaining ratings in attack, defense, magic, and difficulty were aggregated by match – team groupings. The ratings were averaged while the roles were summed up to produce a total number of champions representing each role on the team.

Inter-quartile range outlier detection method was employed on match duration to identify and remove matches that were abnormally long (Chaudhary, 2024). Matches lasting greater than 1964.5 seconds were removed.

Rare role stackings (such as 4 or 5 of one role on a team) were also identified and removed. Any teams that had role stacks represented by less than 50 matches were removed from the dataset.

The final prepared dataset consisted of 11 columns and 34,375 data points:

Feature	Data Type	Description
win	Boolean, Categorical	Team result / response variable; win = True
attack_mean	Float, Continuous	Average champion attack rating
defense_mean	Float, Continuous	Average champion defense rating
magic_mean	Float, Continuous	Average champion magic rating
difficulty_mean	Float, Continuous	Average champion difficulty rating
primary_role_Assassin_sum	Int, Ordinal	Count of Assassin champions
primary_role_Fighter_sum	Int, Ordinal	Count of Fighter champions
primary_role_Mage_sum	Int, Ordinal	Count of Mage champions
primary_role_Marksman_sum	Int, Ordinal	Count of Marksman champions
primary_role_Support_sum	Int, Ordinal	Count of Support champions
primary_role_Tank_sum	Int, Ordinal	Count of Tank champions

Logistic Regression Model

A logistic regression model was chosen for this analysis due to its interpretability and ability to predict a binary outcome (win/loss) (Yasar et.al., 2024). Features were scaled using *StandardScaler()* and a base model was created, yielding a 54.31% accuracy. Collinearity was checked using variance inflation factor (VIF) scores which revealed high correlation between champion ratings (attack, defense, magic) and role counts. To preserve interpretability for players, champion ratings were dropped in favor of role counts.

Recursive feature elimination (RFE) was also employed to further optimize the features in the model (Magedmallek, 2024). The RFE process selected 4 predictors: average difficulty rating and champion counts for Assassin, Mage, and Tank roles. The final model yielded a 54.55% accuracy, with a McFadden's R-squared of 0.008. The model predicted wins better than random chance (50%) but only explained < 1% of the variance in wins.

The logistic regression coefficients suggest that more Mage and Tank champions improves win probability, while the presence of Assassin or higher difficulty champions reduces it.

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RFE-Reduced Logistic Regression Results

Features Included:
Index(['difficulty_mean', 'primary_role_Assassin_sum', 'primary_role_Mage_sum',
      'primary_role_Tank_sum'],
      dtype='object')

Accuracy: 0.5455
Precision: 0.5399
Recall: 0.6242
AUC-ROC: 0.5541

Confusion Matrix:
[[1601 1831]
 [1294 2149]]

Classification Report:

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	precision	recall	f1-score	support
0	0.55	0.47	0.51	3432
1	0.54	0.62	0.58	3443
accuracy			0.55	6875
macro avg	0.55	0.55	0.54	6875
weighted avg	0.55	0.55	0.54	6875

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Log-Likelihood: -4729.461277084867
Null Log-Likelihood: -4765.378066345869
McFadden's R-squared: 0.0075370282821114465
Durbin-Watson statistic: 1.96992


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	Feature	Coefficient	Change in Odds
0	difficulty_mean	-0.067559	0.934672
1	primary_role_Assassin_sum	-0.153414	0.857775
2	primary_role_Mage_sum	0.099063	1.104136
3	primary_role_Tank_sum	0.047809	1.048970

Figure 1: Final logistic regression model results after VIF and RFE reductions.

Random Forest Model

While logistic regression provided interpretability, a random forest model was also implemented to attempt to account for possible non-linear relationships between team composition and match outcomes (IBM, 2024). The baseline model used 500 trees and a minimum leaf sample size of 50 and achieved an accuracy of 55.03%.

The champion ratings were again removed in favor of role counts for better interpretability and actionable insights. The final random forest model yielded an accuracy of 54.89% and AUC-ROC score of 0.57, slightly higher than the logistic regression model. The feature importance scores show the presence of Assassin champions had the strongest influence, following by Mage and Marksman champions.

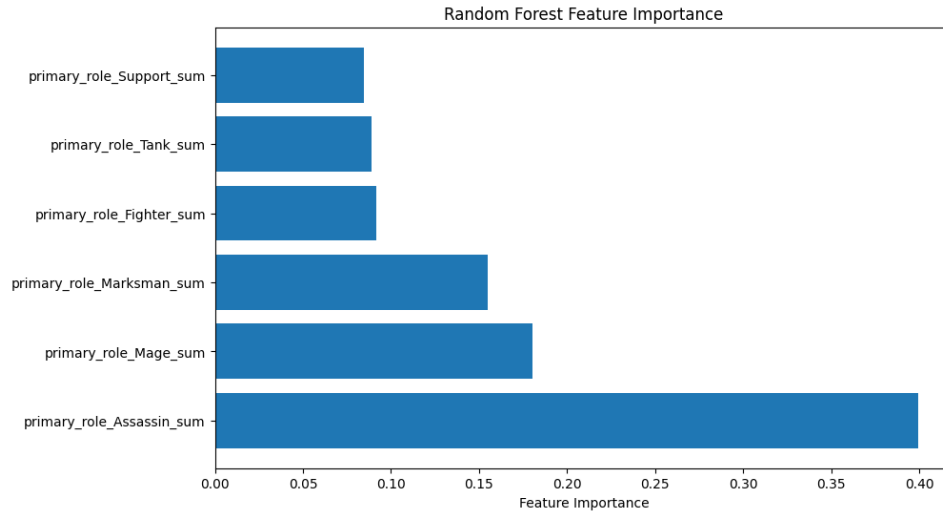


Figure 2: Feature importance of final random forest model.

A sample tree from the model demonstrates how the non-linear interactions were captured and provides key insights into the most optimal team composition:

- A team of 0 Assassins, 2 Mages, 2 Tanks, and 1 Marksman has high win probability.
- Having any Assassins, especially more than 1, dramatically reduces win probability.
- Teams with no Marksmen also struggle to achieve high win probabilities.

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Highest Win Probability: 0.615 (Leaf 87)
Lowest Win Probability: 0.351 (Leaf 119)

Decision Path for Highest Win Probability:
primary_role_Assassin_sum <= 0.50 → primary_role_Mage_sum > 1.50 →
primary_role_Marksman_sum > 0.50 → primary_role_Tank_sum > 0.50 →
primary_role_Mage_sum <= 2.50 → primary_role_Marksman_sum <= 1.50 →
primary_role_Tank_sum > 1.50

Decision Path for Lowest Win Probability:
primary_role_Assassin_sum > 0.50 → primary_role_Mage_sum > 0.50 →
primary_role_Marksman_sum <= 0.50 → primary_role_Fighter_sum > 0.50 →
primary_role_Assassin_sum > 1.50
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Figure 3: Example decision tree showing path for high and low win probabilities based on team composition.

Findings and Proposed Actions

The study confirmed that ARAM match outcomes are not purely dictated by randomness and a predictive model that achieves a win prediction accuracy of > 50% is possible. Certain compositions yield statistically significant advantages and players should take the following actions to ensure a higher chance of winning:

- **Ensure at least one Tank** – Both models showed that at least one Tank champion improves the probability of winning, holding all else equal.
- **Avoid any Assassins** – Having Assassin champions, even 1, had a clear detrimental effect on win probability and should be completely avoided.

- **Prioritize Mages** – The presence of 1 or 2 Mages improved win probability. If the team is already balanced, players should prioritize Mage champions over others.

Despite these insights, there are limitations to this analysis. Player skill levels were not accounted for, meaning a highly skilled player could outperform weaker opponents regardless of team composition. The models also did not consider in-game factors such as item builds, positioning, and player cooperation, which are likely to contribute to match outcomes. The data was also collected from a single game version, meaning balance changes in future patches could alter champion effectiveness.

Future research could benefit from exploring advanced machine learning techniques to improve accuracy and applicability. Neural networks, gradient boosting models, and deep learning approaches could capture more complex patterns in the data, such as champion synergies and dynamic interactions. Additionally, incorporating player skill ratings, item build strategies, and real-time gameplay adaptations into the model could significantly enhance predictive power.

To further translate findings into actionable strategies, the study suggests developing a pre-match decision tool. This tool would allow players to input their assigned champions and receive live recommendations on whether to reroll, swap, or proceed with the given composition. By integrating real-time analysis, this tool could increase ARAM win probability by 5-10% over random selection, or 5 to 10 additional wins per 100 matches.

The expected benefits of this study extend beyond individual matches. By optimizing reroll decisions, players could achieve a higher long-term win rate, making their ARAM experience more rewarding. These insights also contribute to esports analytics, providing a structured approach to understanding the impact of team composition in a game mode often dismissed as purely random. If implemented at scale, the findings could influence competitive ARAM tournaments, where understanding the impact of champion roles could lead to new strategies for structured play.

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